SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

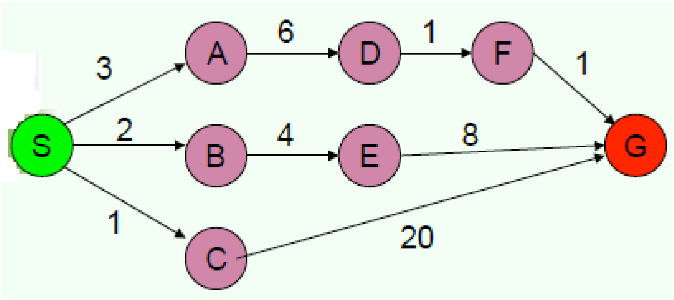
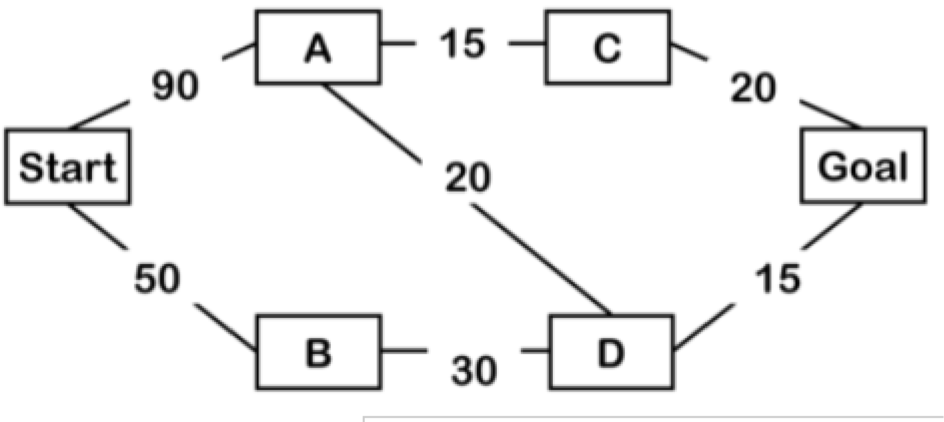
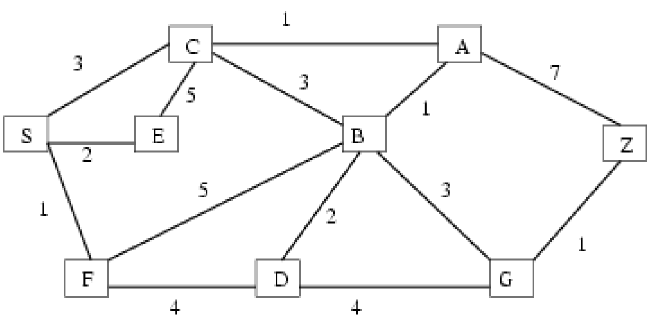
Continuous Assessment Test-1, Fall Semester 2021-2022

CSE3013 – Artificial Integelince

Slot: C1+TC1 **Maximum Marks: 30**

General Instructions (if any): Answer ALL Questions

S.No	Question	Max Marks	CO	BL
1 A	Define in your own words with example: (a) intelligence, (b) artificial intelligence, (c) agent, (d) rationality (e) logical reasoning.	10	CO1	L1
1 B	For each of the following activities, give a PEAS description of the task environment and characterize it in terms of the properties: i) Shopping for used AI books on the Internet. ii) Playing a tennis match	10	CO1	L1
1 C	For each of the following assertions, say whether it is true or false and support your answer with examples or counterexamples where appropriate. a. An agent that senses only partial information about the state cannot be perfectly rational. b. There exist task environments in which no pure reflex agent can behave rationally. c. There exists a task environment in which every agent is rational. d. The input to an agent program is the same as the input to the agent function. e. Suppose an agent selects its action uniformly at random from the set of possible actions. There exists a deterministic task environment in which this agent is rational.	10	CO1	L3
2 A	Give a complete problem formulation for each of the following. Choose a formulation that is precise enough to be implemented. a. Using only four colors, you have to color a planar map in such a way that no two adjacent regions have the same color. b. A 3-foot-tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, movable, climbable 3-foot-high crates	10	CO4	L4

2 B	Illustrate DFS and uniform cost search for the following graph. Compare the search strategies based on the evaluation criteria and write the order of traversal with starting node as “S” and goal node “G”.  	10	CO4	L2
2 C	Illustrate DFS and uniform cost search for the following graph. Compare the search strategies based on the evaluation criteria and write the order of traversal with starting node as “Start” and goal node “Goal”.  		CO4	L2
3 A	Consider the following graph with starting state S and goal state Z. The numbers on the edges indicate the cost of traversing the edge. Steps required (5 +5=10 marks)  	10	CO4	L3

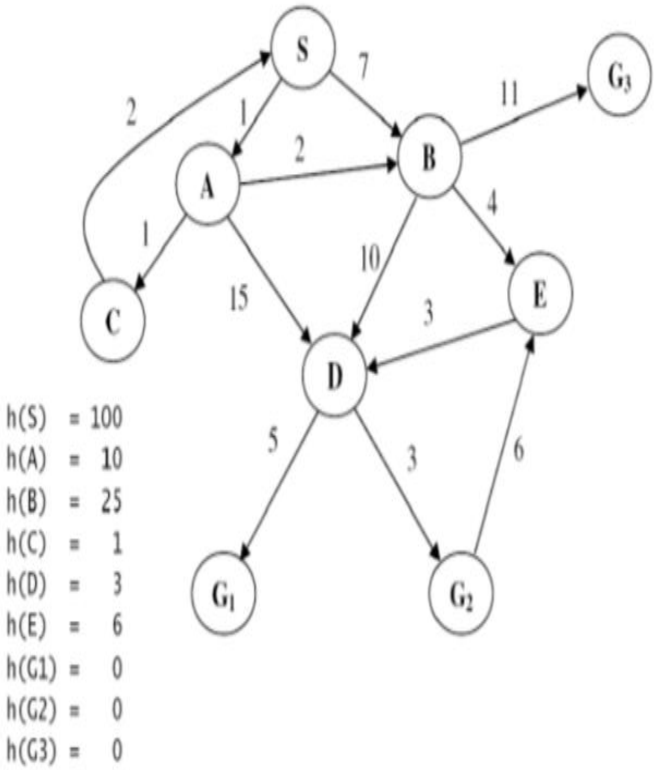
The heuristic values are given below

n	S	F	E	C	D	B	A	G	Z
$h(n)$	8	6	6	5	4	3	2	1	0

(a) Apply Greedy Best first search algorithm and list the nodes in the order in which they are expanded.

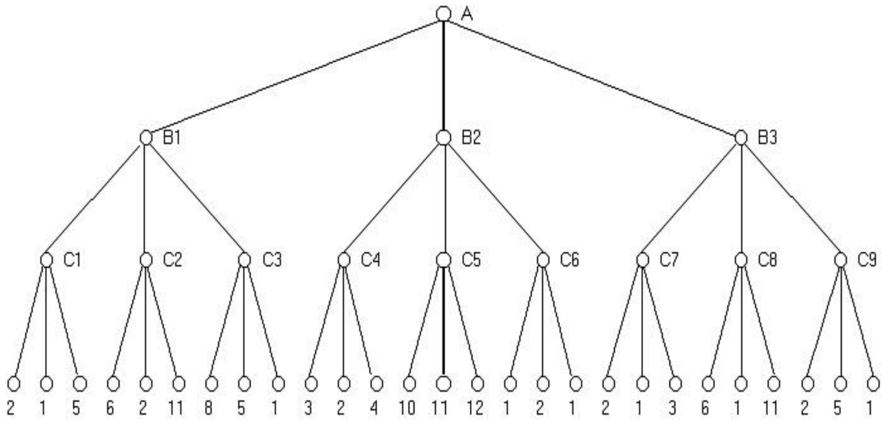
(b) Apply A* search algorithm to the same graph and prove that it returns the optimal path

3 B	Consider the search space below, where S is the start node and G1, G2, and G3 satisfy the goal test. Arcs are labelled with the cost of traversing them and the h function's values are reported beside the graph. For each of the following search strategies, indicate which goal state is reached (if any) and list, in order, all the states popped off of the OPEN list, and CLOSED. When all else is equal, nodes should be removed from OPEN in alphabetical order. (5 +5=10 marks)	10	CO4	L3



(a) Apply Greedy Best first search algorithm and list the nodes in the order in which they are expanded.

(b) Apply A* search algorithm to the same graph and prove that it returns the optimal path

3 C	Write the pseudo code for Alpha- Beta Pruning. Apply both minimax algorithm and alpha beta pruning to the following graph. What are the nodes that are to be pruned? (2+3+5)  	10	CO4	L3